Computer Networks Project (23XW46)

SecureGate- A Network Security Gateway with Real-Time Threat Monitoring

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Abstract:

SecureGate is a Linux-based network security gateway designed to protect user devices from untrusted public Wi-Fi networks. The system creates an isolated network layer between client devices and potentially malicious public networks, functioning as an intelligent router with integrated security monitoring and threat detection capabilities.

Network traffic is routed through a secure gateway layer implementing NAT translation, DHCP address assignment, and firewall-based packet filtering. Real-time monitoring automatically tracks connected devices, analyzes DNS queries, and detects suspicious network behavior including port scanning and abnormal traffic patterns using packet-level inspection.

All security events are logged and analyzed through a custom Python-based monitoring module. The system provides immediate alerts for detected threats while maintaining transparent internet connectivity for legitimate traffic, demonstrating practical implementation of routing protocols, network address translation, and intrusion detection mechanisms.

Key Features

- Automated hotspot creation with isolated network architecture
- NAT-based internet sharing with routing table management
- Rule-based firewall with customizable security policies
- DNS filtering and malicious domain blocking
- Real-time port scan detection and alerting
- Traffic analysis with packet-level inspection (Scapy)
- Connection logging and device tracking

Tech Stack

- **Network Layer**: Linux networking stack (hostapd, dnsmasq, iptables)
- **Monitoring**: Python 3 with Scapy library
- **Analysis Tools**: Wireshark, tcpdump
- **Simulation**: Cisco Packet Tracer
- **Platform**: Linux (Ubuntu/Debian-based distribution.)